

The Balance of the Count:

One Energy in the Prime Counts and in the Sawtooth, and the Riemann Hypothesis in
Real Variables

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Abstract

We write the Riemann Hypothesis (RH) in real variables and measure the constant it predicts in three independent ways, which agree to within 2%. Every integer $k > 1$ has as many squarefree divisors with an even number of prime factors as with an odd number, and this pairing gives the rebuild law $\sum_{n \leq x} M(x/n) = 1$ for the Mertens function M : the balance at every scale is fixed by the balances at all smaller scales. We prove that the Möbius-weighted sawtooths rebuild the constant exactly, $-\sum_{k \geq 1} \mu(k) \{1/(kt)\} = \chi_{(0,1]}(t)$ for every $t > 0$, and that the first N of them fall short by a value term $x m(N)$, $m(N) = \sum_{k \leq N} \mu(k)/k$, and by a signed sum $R_N(x)$ of the unfinished divisor sums $c_N(n)$ over $N < n \leq x$; the integral of the squared shortfall over $[1/N, 1]$ is exactly $(N-1) m(N)^2$, which does not tend to 0. We prove that RH is equivalent to the statement that the energy $E(X) = \int_1^X ((\psi(x) - x)/x)^2 dx$ of the prime-count error grows more slowly than every power of X . Measured in doublings $u = \log_2 x$, the error of the prime counts to 10^{10} shows real waves at 1.560, 2.320, 2.760, 3.358, 3.633, 4.148, 4.515 and 4.780 turns per doubling; each wave grows by $2^{u(1/2+\delta)}$ with $|\delta| \leq 0.0022$ between the two halves of the range, and the first wave's amplitude in the two halves agrees to 0.4%. The mean energy per unit of $\log x$ is 0.04592 over $2^{12} \leq x \leq 10^{10}$, and 0.04587 and 0.04597 in the two halves. The sum $\sum 1/|\rho|^2$ over the zeros with $|\gamma| < 3000$ (2469 conjugate pairs), with the density tail, is 0.046192, and the sawtooth distance of Nyman, Beurling and Báez-Duarte, computed from exact integrals of fractional parts with no reference to primes, gives $d_N^2 \log N$ between 0.0454 and 0.0470 for $40 \leq N \leq 60$; all three agree with $2 + \gamma - \log 4\pi = 0.0461914$ to within 2%. The rebuild law, with a table of μ to 10^9 , reproduces $M(10^n)$ exactly for $9 \leq n \leq 15$ and gives $|M(x)| \leq 0.424\sqrt{x}$ at every doubling from 2^{34} to 2^{49} . The integers free of the primes up to 37 carry no wave (amplitude at most 0.0018 against 0.14–0.18), while at $x = 10^{10}$ the Möbius sum over the integers whose prime factors are at most $x/2$ is $+2200.6\sqrt{x}$ and the primes in $(x/2, x]$ bring it to $M(x) = -0.337\sqrt{x}$. Generalized-integer systems of Diamond, Montgomery and Vorhauer and of Broucke, Debruyne and Révész obey the same pairing and the same rebuild law, the latter with integer-count error $O(x^{1/2+\epsilon})$, and have prime errors far above $\sqrt{x}$; a proof of RH must therefore use more about the integer count than these properties, and the ordinary count has more: it is exact, $\lfloor x \rfloor = x - \{x\}$ with $0 \leq \{x\} < 1$. We state the remaining step as the subpolynomial growth of the energy.

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1 Introduction

Let μ be the Möbius function, $M(x) = \sum_{n \leq x} \mu(n)$ the Mertens function, Λ the von Mangoldt function and $\psi(x) = \sum_{n \leq x} \Lambda(n)$. The Riemann Hypothesis (RH) is the conjecture that every nontrivial zero $\rho = \beta + i\gamma$ of ζ has $\beta = \frac{1}{2}$. Littlewood proved that RH is equivalent to $M(x) \ll x^{1/2+\varepsilon}$ for every $\varepsilon > 0$ [18, §14.25], and von Koch proved that under RH $\psi(x) - x \ll \sqrt{x} \log^2 x$ [20]. Both statements compare a count with a balanced walk: a walk of n independent steps ± 1 has mean square n , because every cross term $s_i s_j$ is cancelled by its opposite, and so spreads as $\sqrt{n}$. RH says that the Möbius sign $\mu(n)$, the parity of the number of prime factors of a squarefree n , summed over the integers, spreads no faster than such a walk, up to x^ε .

The main criterion (Theorem 1.3) and every measurement in this paper are stated in real variables. The zeros of ζ enter the measurements only through the real waves they produce in the counts, each wave being described by two real numbers, its growth exponent β and its pitch, and the constants π and e enter only as units: we measure x in doublings, $u = \log_2 x$, and pitch in turns per doubling.

1.1 Main results

- Theorem 1.1** (pair cancellation and the rebuild law). (i) *For every integer $k > 1$, $\sum_{d|k} \mu(d) = 0$: the squarefree divisors of k fall into pairs $\{d, dp\}$, $p \mid k$ a fixed prime, with $\mu(dp) = -\mu(d)$.*
- (ii) *For every real $x \geq 1$, $\sum_{n \leq x} M(x/n) = 1$.*
- (iii) *M is the only function $\bar{F}$ on $(0, \infty)$ with $F = 0$ on $(0, 1)$ and $\sum_{n \leq x} F(x/n) = 1$ for all $x \geq 1$.*

The law determines the balance at every scale from the balances at smaller scales. The next theorem splits a sum of dilated sawtooths into a value part and a count part.

Theorem 1.2 (count and value of the sawtooth sum). *For an integer $N \geq 1$ and real $x \geq 1$ let $m(N) = \sum_{k \leq N} \mu(k)/k$, $c_N(n) = \sum_{k|n, k \leq N} \mu(k)$ and $R_N(x) = \sum_{N < n \leq x} c_N(n)$. Then*

$$\sum_{k \leq N} \mu(k) \{x/k\} = x m(N) - 1 - R_N(x), \quad (1)$$

and $R_N(x) = 0$ for $x < N + 1$. Consequently:

- (a) for every $t > 0$, $-\sum_{k \geq 1} \mu(k) \{1/(kt)\} = \chi_{(0,1]}(t)$, the series converging for each t ;
- (b) with $\varsigma_k(t) = \{1/(kt)\}$, the truncation satisfies $\chi_{(0,1]}(t) + \sum_{k \leq N} \mu(k) \varsigma_k(t) = m(N)/t$ for $1/N \leq t \leq 1$, and

$$\int_{1/N}^1 \left| \chi_{(0,1]}(t) + \sum_{k \leq N} \mu(k) \varsigma_k(t) \right|^2 dt = (N-1) m(N)^2.$$

The constant function is rebuilt exactly, pointwise, by the pair-weighted sawtooths; a finite rebuild falls short on $[1/N, 1]$ only by the value imbalance $m(N)$ of the first N pairs, and beyond by the integers above N whose divisor pairs are unfinished. Our sieve gives $\sqrt{N} m(N) = +0.311, +0.140, -0.208, -0.154, +0.201, +0.321$ and $+0.205$ at $N = 10^2, \dots, 10^8$: the value imbalance itself sits at the scale $N^{-1/2}$ and is not $o(N^{-1/2})$ (Section 3), so the unweighted rebuild does not converge in mean square, and the rate of every rebuild is again a question of balance.

Theorem 1.3 (the energy form of RH). *Let $E(X) = \int_1^X ((\psi(x) - x)/x)^2 dx$. Then RH holds if and only if $E(X) \ll_\varepsilon X^\varepsilon$ for every $\varepsilon > 0$. Under RH, $E(X) \sim c \log X$ with $c = \sum m(\rho)^2 |\rho|^{-2}$ over the distinct zeros, $m(\rho)$ the multiplicity [7], so that $c = \sum_\rho |\rho|^{-2}$ when the zeros are simple; and $\sum_\rho 1/(\rho(1-\rho)) = 2 + \gamma - \log 4\pi = 0.0461914\dots$ holds unconditionally, the sum being $\sum_\rho |\rho|^{-2}$ when every zero lies on the line.*

In doublings, $E(2^U)/(U \log 2)$ is the mean over $0 \leq u \leq U$ of $((\psi(x) - x)/\sqrt{x})^2$: RH says that the energy of the count's error per doubling stays bounded, up to growth slower than every power of x . A zero with $\beta > \frac{1}{2}$ would make $E(X)$ exceed $X^{2\beta-1-\varepsilon}$ for arbitrarily large X , for every $\varepsilon > 0$ (Section 5).

Theorem 1.4 (Diamond, Montgomery and Vorhauer [10]). *There is a system of Beurling generalized primes whose integers satisfy $N_B(x) = \kappa x + O(x^\theta)$ with $\kappa > 0$ and $\frac{1}{2} < \theta < 1$, whose primes satisfy $\pi_B(x) = \text{li}(x) + O(x \exp(-c\sqrt{\log x}))$, and whose zeta function has infinitely many zeros on the curve $\sigma = 1 - a/\log t$, $t \geq 2$.*

Every Beurling system has unique factorisation into its generalized primes, so Theorem 1.1(i) and (ii) hold in it verbatim, with the sums taken over its integers; the dual form $\sum_{n \leq x} \mu(n) \lfloor x/n \rfloor = 1$ becomes $\sum_{n \leq x} \mu_B(n) N_B(x/n) = 1$. Broucke, Debruyne and Révész [5] construct systems whose integer-count error has any exponent $\beta \in [\frac{1}{2}, 1)$ and whose prime-count error has any exponent $\alpha \in [0, 1)$. Hence (Theorem 8.1) no argument that uses only the pairing, the rebuild law and a bound $O(x^{1/2+\varepsilon})$ for the error of the integer count can prove RH: a proof must use more about the ordinary count, which is exact, $\lfloor x \rfloor = x - \{x\}$ with $0 \leq \{x\} < 1$.

1.2 Main computations

All numbers below are computed from exact integer counts, from the zeros of ζ below 3000 [16], and from exact integrals of fractional parts, by the methods of Section 9.

- (1) *Waves in doublings* (Section 4, Fig. 1). The spectra of $M(x)/\sqrt{x}$ and $(\psi(x) - x)/\sqrt{x}$ over $12 \leq u \leq 33.22$ have peaks at 1.560, 2.320, 2.760, 3.358, 3.633, 4.148, 4.515, 4.780, 5.295, 5.490 and 5.843 turns per doubling; the pitches $\gamma_k \log 2/2\pi$ of the first eleven zeros are 1.5593, 2.3191, 2.7591, 3.3564, 3.6333, 4.1464, 4.5141, 4.7797, 5.2958, 5.4909 and 5.8436. The first wave of $(\psi - x)/\sqrt{x}$ has amplitude 0.1420 in $12 \leq u \leq 22.6$ and 0.1414 in $22.6 \leq u \leq 33.22$, against $2/|\rho_1| = 0.14141$; the growth exponents $\frac{1}{2} + \delta$ of the first eight waves satisfy $|\delta| \leq 0.0022$.

- (2) *One energy* (Section 5, Fig. 2). The mean of $((\psi - x)/\sqrt{x})^2$ over $12 \leq u \leq 33.22$ is 0.04592, and 0.04587 and 0.04597 over the two halves. The sum $\sum 2/(\frac{1}{4} + \gamma^2)$ over the 2469 zeros with $0 < \gamma < 3000$ is 0.045431, and with the density tail 0.046192. The sawtooth distance gives $d_N^2 \log N$ between 0.0454 and 0.0470 for $40 \leq N \leq 60$. The constant is $2 + \gamma - \log 4\pi = 0.0461914$.
- (3) *The rebuild law to 10^{15}* (Section 7). The law reproduces $M(10^n)$ for $9 \leq n \leq 15$ exactly, $M(10^{15}) = -3,216,373$, and gives $\max |M(x)|/\sqrt{x} = 0.424$ over the doublings $x = 2^{34}, \dots, 2^{49}$ (Table 4).
- (4) *What the small primes carry* (Section 8). The count of integers up to x free of the primes up to 37 differs from $x \prod_{p \leq 37} (1 - 1/p)$ by at most 38.9 on the grid of Section 4 ($2^{12} \leq x \leq 10^{10}$) and carries no wave: its amplitude at every pitch in $[0.3, 6]$ is at most 0.0018. At $x = 10^{10}$ the squarefree integers whose prime factors are at most $x/2$ give $M(x, x/2) = 220,064,566 = 2200.6\sqrt{x}$; the primes in $(x/2, x]$ bring it to $M(x) = -33,722 = -0.337\sqrt{x}$. The sum $L(x)$ of the Liouville function is at most 0 for $2 \leq x < 906,150,257$ and $L(906,150,257) = 1$ [17].

1.3 The remaining step

By Theorem 1.3, RH is equivalent to the following statement about the counts alone.

Conjecture 1 (subpolynomial energy of the count). *For every $\varepsilon > 0$, $\int_1^X ((\psi(x) - x)/x)^2 dx \ll_\varepsilon X^\varepsilon$.*

Theorem 1.4 and Theorem 8.1 show what a proof of Conjecture 1 cannot avoid: the pairing and the rebuild law hold in systems whose energy exceeds a power of X for arbitrarily large X , so a proof must use more about the ordinary count than these, such as its exactness. The criterion of Nyman and Beurling [4], in the form of Báez-Duarte [2], is the statement in which the exact count appears alone: RH holds if and only if the distance d_N of Section 6 tends to 0, and Theorem 1.2 shows that the pair weights $-\mu(k)$ rebuild the target exactly and fall short, at every finite N , by the value imbalance $m(N)$ and the unfinished pairs R_N .

Relation to prior work. Littlewood's criterion, von Koch's bound, Cramér's mean value theorem, the identity $\sum_\rho 1/(\rho(1-\rho)) = 2 + \gamma - \log 4\pi$, the Nyman–Beurling criterion with Báez-Duarte's strengthening, Burnol's lower bound and the asymptotic of Bettin, Conrey and Farmer are classical or published and are credited where they are used. The pairing of Theorem 1.1(i) is the classical proof of $\sum_{d|k} \mu(d) = 0$, and (ii) is the classical identity for M . The implication from subpolynomial energy to RH in Section 5 is a Mellin argument of Landau type, included for completeness. Closed forms for the Gram integrals of Section 6 are due to Vasyunin [19], and d_N was computed by Landreau and Richard [8]; we recompute both by exact piecewise integration. The values of M in Section 7 reproduce published values [13, 14]. New here are the count–value decomposition (1) with its exact shortfall; the measurement of the waves and of the energy in doublings from exact counts, and its comparison with the zeros and with $d_N^2 \log N$; and the computations of Section 8.

Methods of proof. Theorem 1.1 is a pairing argument and an interchange of summation. Theorem 1.2 writes each floor as a value minus a sawtooth and counts which divisor pairs close below N . The implication from subpolynomial energy to RH uses Cauchy–Schwarz on dyadic blocks to make the Mellin transform of $\psi(x) - x$ absolutely convergent on $\text{Re } s > \frac{1}{2}$, and the identity $\int_1^\infty \psi(x)x^{-s-1} dx = -\zeta'(s)/(s\zeta(s))$; the converse is von Koch's bound.

Organisation. Section 2 proves Theorem 1.1; Section 3 proves Theorem 1.2; Section 4 gives the real form of the explicit formula and the measured waves; Section 5 proves Theorem 1.3

and measures the energy; Section 6 computes the sawtooth distance; Section 7 evaluates the rebuild law to 10^{15} ; Section 8 treats the small primes and the newest ones; Section 9 states the methods of computation.

2 Pairs and the rebuild law

Proof of Theorem 1.1. (i) Let $k > 1$ and let p be a prime factor of k . A divisor d of k that is not squarefree has $\mu(d) = 0$. The squarefree divisors not divisible by p and those divisible by p are in bijection by $d \mapsto dp$, and $\mu(dp) = -\mu(d)$ because dp has one more prime factor. The sum therefore vanishes.

(ii) For $x \geq 1$,

$$\sum_{n \leq x} M(x/n) = \sum_{n \leq x} \sum_{m \leq x/n} \mu(m) = \sum_{mn \leq x} \mu(m) = \sum_{k \leq x} \sum_{m|k} \mu(m) = 1,$$

by (i), since only $k = 1$ contributes.

(iii) Let F satisfy the hypotheses and put $D = F - M$. Then $D = 0$ on $(0, 1)$ and $\sum_{n \leq x} D(x/n) = 0$ for $x \geq 1$, so $D(x) = -\sum_{2 \leq n \leq x} D(x/n)$. If $D = 0$ on $(0, 2^j)$ for some $j \geq 0$, then for $x < 2^{j+1}$ every argument x/n with $n \geq 2$ is below 2^j , so $D(x) = 0$. By induction $D \equiv 0$. $\square$

Remark 2.1. Grouping the terms of $M(v) = 1 - \sum_{2 \leq n \leq v} M(\lfloor v/n \rfloor)$ by the value of $\lfloor v/n \rfloor$ gives, for $v \geq 2$ and with $q = \lfloor \sqrt{v} \rfloor$,

$$M(v) = 1 - \sum_{2 \leq n \leq v/(q+1)} M(\lfloor v/n \rfloor) - \sum_{t=1}^q \left(\left\lfloor \frac{v}{t} \right\rfloor - \left\lfloor \frac{v}{t+1} \right\rfloor \right) M(t),$$

in which every argument is either at most $\sqrt{v}$ or of the form $\lfloor x/k' \rfloor$ with $k' > k$ when $v = \lfloor x/k \rfloor$. Evaluated on the values $\lfloor x/k \rfloor$ in increasing order, with M on $[1, L]$ from a table, this computes $M(x)$ in $O(x/\sqrt{L})$ operations; Section 7 uses $L = 10^9$.

3 Count and value

Proof of Theorem 1.2. For $x \geq 1$,

$$\sum_{k \leq N} \mu(k) \left\lfloor \frac{x}{k} \right\rfloor = \sum_{k \leq N} \mu(k) \sum_{n \leq x, k|n} 1 = \sum_{n \leq x} c_N(n).$$

For $n \leq N$ every divisor of n is at most N , so $c_N(n) = \sum_{k|n} \mu(k)$, which is 1 for $n = 1$ and 0 for $1 < n \leq N$ by Theorem 1.1(i). Hence the sum is $1 + R_N(x)$, and $R_N(x) = 0$ when no integer lies in $(N, x]$, that is, for $x < N + 1$. Writing $\{x/k\} = x/k - \lfloor x/k \rfloor$ gives (1).

(a) Let $t > 0$ and $x = 1/t$. If $x \geq 1$, take $N \geq x$; then $R_N(x) = 0$ and (1) reads $\sum_{k \leq N} \mu(k) \{x/k\} = x m(N) - 1$. Since $\sum_{k \geq 1} \mu(k)/k = 0$ [1, Ch. 4], the partial sums tend to -1 . If $x < 1$, then $\{x/k\} = x/k$ for all k , and the partial sums are $x m(N) \rightarrow 0$.

(b) For $1/N \leq t \leq 1$ we have $1 \leq x \leq N < N + 1$, so $\chi_{(0,1]}(t) + \sum_{k \leq N} \mu(k) \zeta_k(t) = 1 + x m(N) - 1 = m(N)/t$. With $dt = dx/x^2$,

$$\int_{1/N}^1 \frac{m(N)^2}{t^2} dt = m(N)^2 \int_1^N dx = (N - 1) m(N)^2.$$

$\square$

N	$m(N)$	$\sqrt{N} m(N)$	$(N-1)m(N)^2$
10^2	$+3.113 \times 10^{-2}$	+0.3113	0.0959
10^3	$+4.412 \times 10^{-3}$	+0.1395	0.0194
10^4	-2.083×10^{-3}	-0.2083	0.0434
10^5	-4.872×10^{-4}	-0.1541	0.0237
10^6	$+2.006 \times 10^{-4}$	+0.2006	0.0402
10^7	$+1.015 \times 10^{-4}$	+0.3210	0.1031
10^8	$+2.047 \times 10^{-5}$	+0.2047	0.0419

Table 1: The value imbalance $m(N) = \sum_{k \leq N} \mu(k)/k$ and the exact shortfall of the unweighted rebuild on $[1/N, 1]$ (Theorem 1.2(b)).

The shortfall on $[1/N, 1]$ does not tend to 0. By partial summation $M(N) = Nm(N) - \sum_{k < N} m(k)$, so $m(N) = o(N^{-1/2})$ would give $M(N) = o(\sqrt{N})$, whereas $\limsup_{x \rightarrow \infty} |M(x)|/\sqrt{x} > 1$ [11]; hence $\limsup_N (N-1)m(N)^2 > 0$, and the L^2 distance of $-\sum_{k \leq N} \mu(k)\zeta_k$ from $\chi_{(0,1]}$ does not tend to 0. Table 1 shows $m(N)$ at the order $N^{-1/2}$, the scale of a balanced walk. By partial summation and $\sum_k \mu(k)/k = 0$, $m(N) = M(N)/N - \int_N^\infty M(t)t^{-2} dt$, so Littlewood's criterion gives $m(N) \ll N^{-1/2+\varepsilon}$ under RH, and the values in Table 1 are of that size: $|\sqrt{N} m(N)|$ lies between 0.139 and 0.321. A rebuild whose shortfall tends to 0 needs smoothed weights; the optimal ones are the subject of Section 6.

4 Waves in doublings

Lemma 4.1 (the explicit formula in real form). *For $x > 1$ not a prime power,*

$$\psi(x) - x = - \sum_{\gamma > 0} \frac{2x^\beta}{|\rho|} \cos(\gamma \log x - \arg \rho) - \log 2\pi - \frac{1}{2} \log(1 - x^{-2}),$$

the sum over the zeros $\rho = \beta + i\gamma$ with $\gamma > 0$ taken in order of increasing γ . With $x = 2^u$, each term is a real wave of amplitude $2^{\beta u} \cdot 2/|\rho|$ and pitch $p = \gamma \log 2/2\pi$ turns per doubling.

Proof. The explicit formula $\psi(x) = x - \sum_\rho x^\rho/\rho - \log 2\pi - \frac{1}{2} \log(1 - x^{-2})$, with the sum over $|\gamma| < T$ and $T \rightarrow \infty$, is classical [9, Ch. 17]. The zeros are symmetric under $\rho \mapsto \bar{\rho}$, and $x^\rho/\rho + x^{\bar{\rho}}/\bar{\rho} = 2 \operatorname{Re}(x^\rho/\rho) = 2x^\beta \cos(\gamma \log x - \arg \rho)/|\rho|$. With $x = 2^u$, $\gamma \log x = 2\pi \cdot (\gamma \log 2/2\pi) u$. $\square$

Each pair of conjugate zeros is thus one real wave, fixed by two real numbers: its growth exponent β and its pitch p . RH is the statement that every wave grows as $2^{u/2} = \sqrt{x}$.

k	$\gamma_k \log 2/2\pi$	peak, M	peak, ψ	δ , M	δ , ψ
1	1.5593	1.560	1.560	-0.0003	-0.0006
2	2.3191	2.320	2.320	-0.0007	-0.0013
3	2.7591	2.760	2.760	-0.0008	+0.0012
4	3.3564	3.358	3.355	+0.0003	+0.0001
5	3.6333	3.633	3.633	-0.0005	-0.0022
6	4.1464	4.148	4.148	-0.0003	+0.0009
7	4.5141	4.515	4.515	-0.0018	-0.0013
8	4.7797	4.780	4.780	-0.0012	-0.0010

Table 2: Pitches of the first eight waves from the counts (spectral grid 0.0025) against $\gamma_k \log 2/2\pi$, and the growth exponents $\frac{1}{2} + \delta$ from the amplitudes over $12 \leq u \leq 22.6$ and $22.6 \leq u \leq 33.22$.

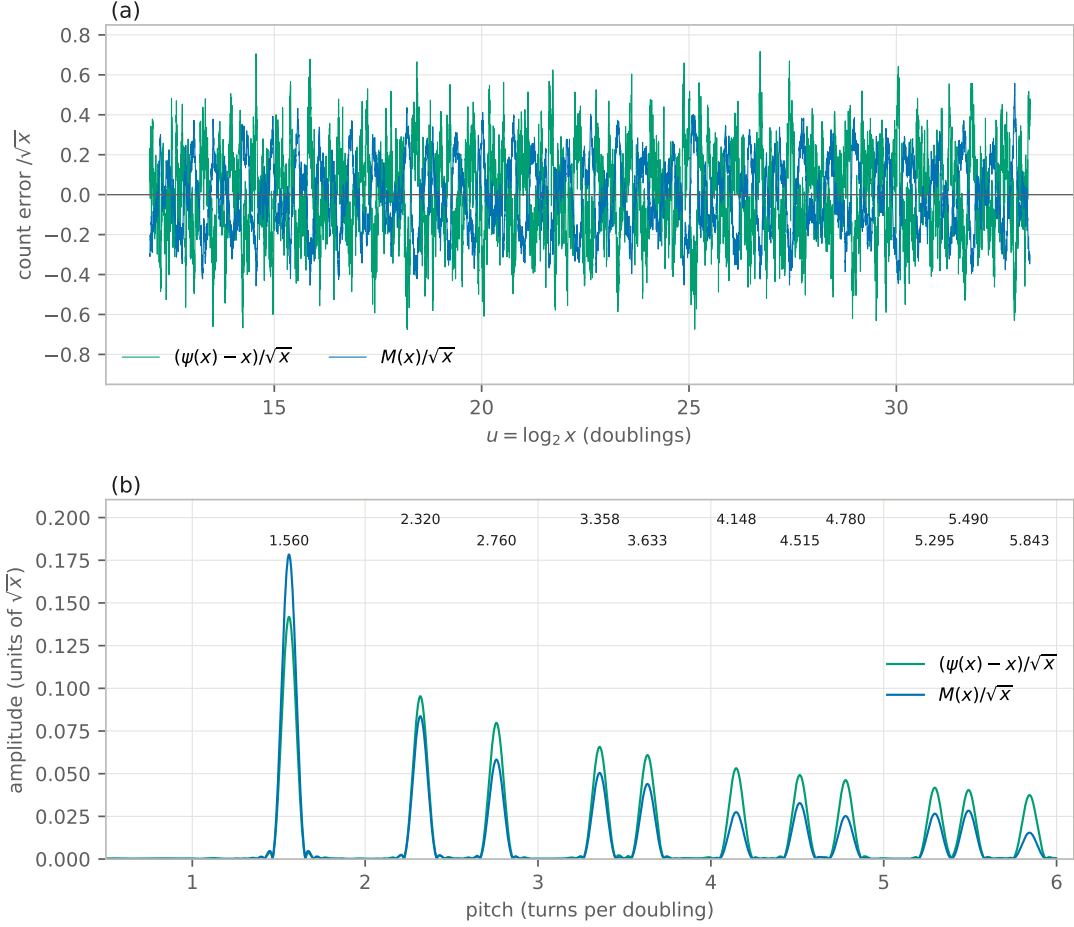


Figure 1: (a) $M(x)/\sqrt{x}$ and $(\psi(x) - x)/\sqrt{x}$ from exact counts, $2^{12} \leq x \leq 10^{10}$, against $u = \log_2 x$. (b) Their spectra in turns per doubling (Hann window over $12 \leq u \leq 33.22$), with the peak positions.

Measurement. From the exact counts at 675,247 checkpoints (Section 9) we interpolate $M(x)$ and $\psi(x) - x$ on the grid $u = 12, 12.001, \dots, 33.219$, divide by $\sqrt{x}$, and compute the Hann-windowed spectrum on pitches $0.3 \leq p \leq 6$ in steps of 0.0025 (Fig. 1). The peaks of Table 2 coincide with $\gamma_k \log 2/2\pi$ to the grid resolution. The means of $M/\sqrt{x}$ and $(\psi - x)/\sqrt{x}$ over the range are -0.0046 and -0.0026 . For the growth exponent we compute the amplitude A_1 of each wave over $12 \leq u \leq 22.6$ and A_2 over $22.6 \leq u \leq 33.22$ at the pitch $\gamma_k \log 2/2\pi$ and put $\delta = \log_2(A_2/A_1)/\Delta u$, $\Delta u = 10.6$. The first wave of $(\psi - x)/\sqrt{x}$ has $A_1 = 0.1420$ and $A_2 = 0.1414$, against $2/|\rho_1| = 0.14141$ from Theorem 4.1; that of $M/\sqrt{x}$ has $A_1 = 0.1786$ and $A_2 = 0.1783$. Beyond the eight waves of Table 2, both spectra have peaks at 5.295, 5.490 and 5.843, next to the pitches 5.2958, 5.4909 and 5.8436 of the ninth to eleventh zeros (Fig. 1b).

5 The energy

Proof of Theorem 1.3. Suppose RH. By von Koch's theorem [20] $|\psi(x) - x| \ll \sqrt{x} \log^2 x$, so the integrand of E is $\ll (\log^4 x)/x$ and $E(X) \ll \log^5 X \ll_\varepsilon X^\varepsilon$.

Conversely, suppose $E(X) \ll_\varepsilon X^\varepsilon$ for every $\varepsilon > 0$, and let $\sigma > \frac{1}{2}$. Choose $0 < \varepsilon < 2\sigma - 1$.

For $j \geq 0$, by Cauchy–Schwarz,

$$\begin{aligned} \int_{2^j}^{2^{j+1}} \frac{|\psi(x) - x|}{x^{\sigma+1}} dx &= \int_{2^j}^{2^{j+1}} \frac{|\psi(x) - x|}{x} x^{-\sigma} dx \\ &\leq E(2^{j+1})^{1/2} \left(\int_{2^j}^{2^{j+1}} x^{-2\sigma} dx \right)^{1/2} \ll 2^{j(\varepsilon+1-2\sigma)/2}, \end{aligned}$$

and the sum over j converges. Hence $F(s) = \int_1^\infty (\psi(x) - x)x^{-s-1} dx$ converges absolutely and is analytic on $\operatorname{Re} s > \frac{1}{2}$. For $\operatorname{Re} s > 1$, partial summation gives $\int_1^\infty \psi(x)x^{-s-1} dx = -\zeta'(s)/(s\zeta(s))$ [9, Ch. 17], and $\int_1^\infty x^{-s} dx = 1/(s-1)$, so

$$-\frac{\zeta'(s)}{\zeta(s)} = s \left(F(s) + \frac{1}{s-1} \right).$$

The right side is meromorphic on $\operatorname{Re} s > \frac{1}{2}$ with a single pole, at $s = 1$. By analytic continuation ζ'/ζ has no pole on $\operatorname{Re} s > \frac{1}{2}$ other than $s = 1$, so ζ has no zero there. By the functional equation there is none on $\operatorname{Re} s < \frac{1}{2}$ in the critical strip either, and every nontrivial zero lies on the line.

Cramér [7] proved under RH that $\int_2^X (\psi(x) - x)^2 x^{-2} dx \sim c \log X$, $c = \sum m(\rho)^2 |\rho|^{-2}$ over the distinct zeros, and $\int_1^2 (\psi(x) - x)^2 x^{-2} dx = 1$. The identity $\sum_\rho \operatorname{Re}(1/\rho) = 1 + \frac{\gamma}{2} - \frac{1}{2} \log 4\pi$ [9, Ch. 12], with $1/\rho + 1/(1-\rho) = 1/(\rho(1-\rho))$ and the symmetries $\rho \mapsto 1-\rho$ and $\rho \mapsto \bar{\rho}$ of the zero set, gives $\sum_\rho 1/(\rho(1-\rho)) = 2 + \gamma - \log 4\pi$; when $\beta = \frac{1}{2}$, $\rho(1-\rho) = |\rho|^2$. $\square$

In the variable u , $dx/x = \log 2 du$ and $\log X = U \log 2$, so $E(2^U)/(U \log 2)$ is the mean over $0 \leq u \leq U$ of $((\psi(x) - x)/\sqrt{x})^2$. By Theorem 4.1 and the orthogonality of waves of different pitch over long ranges of u , the mean square of a sum of waves of constant amplitude $2/|\rho|$, one for each simple zero with $\gamma > 0$, is $\sum_{\gamma>0} \frac{1}{2} (2/|\rho|)^2 = \sum_\rho |\rho|^{-2}$: the energy is the sum of the energies of the waves, and it is constant in u exactly when every amplitude is.

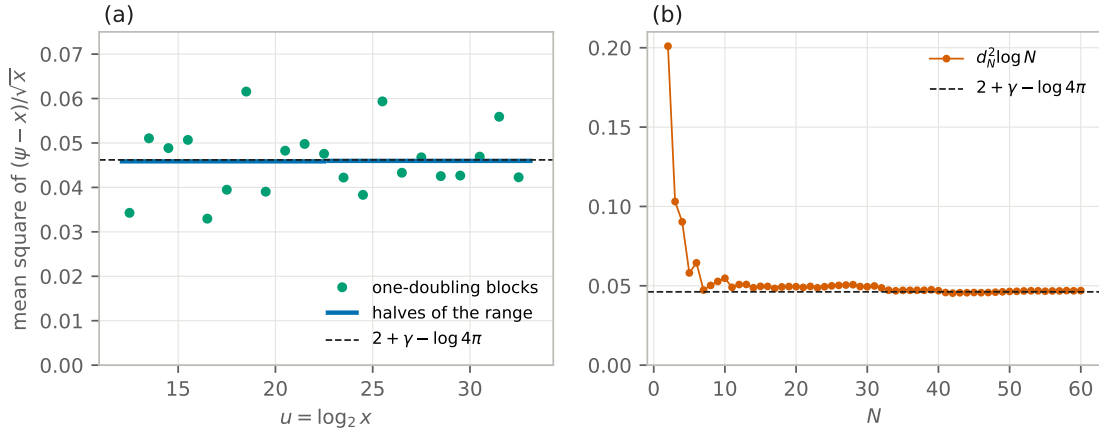


Figure 2: (a) The mean of $((\psi(x) - x)/\sqrt{x})^2$ over blocks of one doubling and over the two halves of $12 \leq u \leq 33.22$, with $2 + \gamma - \log 4\pi$ (dashed). (b) The sawtooth distance $d_N^2 \log N$ for $2 \leq N \leq 60$, with the same constant.

Measurement. Over $12 \leq u \leq 33.22$ the mean of $((\psi - x)/\sqrt{x})^2$ is 0.04592; over the halves it is 0.04587 and 0.04597, and over blocks of one doubling it ranges from 0.033 to 0.062, as a finite window of a sum of waves must (Fig. 2a). Over the 2469 zeros with $0 < \gamma < 3000$, all on the line [15], $\sum 2/(\frac{1}{4} + \gamma^2) = 0.045431$; the zeros above height T , counted by the density $\frac{1}{2\pi} \log \frac{t}{2\pi}$, add $2 \int_T^\infty \frac{1}{2\pi} \log \frac{t}{2\pi} t^{-2} dt = (\log \frac{T}{2\pi} + 1)/(\pi T) = 0.000761$ at $T = 2999.5$, for a total

of 0.046192 against $2 + \gamma - \log 4\pi = 0.0461914$. The energy measured in the counts to 10^{10} is below the energy of the waves by 0.6%, and its values in the two halves of the range differ by 0.2%.

6 The sawtooth rebuilds the constant

Let $\varsigma_k(t) = \{1/(kt)\}$ and $\chi = \chi_{(0,1]}$, and for $N \geq 1$ let

$$d_N^2 = \min_{c \in \mathbb{R}^N} \int_0^\infty \left| \chi(t) - \sum_{k \leq N} c_k \varsigma_k(t) \right|^2 dt.$$

Báez-Duarte [2], strengthening the criterion of Nyman and Beurling [4], proved that RH holds if and only if $d_N \rightarrow 0$. Burnol [6] proved unconditionally that $\liminf_N d_N^2 \log N \geq \sum_{\text{Re } \rho=1/2} m(\rho)^2/|\rho|^2$, $m(\rho)$ the multiplicity, and Bettin, Conrey and Farmer [3] proved $d_N^2 \sim (2 + \gamma - \log 4\pi)/\log N$ under RH and a hypothesis on $\sum 1/|\zeta'(\rho)|^2$. By Theorem 1.2(a) the weights $c_k = -\mu(k)$ rebuild χ pointwise; d_N measures how well N sawtooths can do it in mean square.

Lemma 6.1. *With $u = 1/t$, $\langle \varsigma_j, \varsigma_k \rangle = \int_0^\infty \{u/j\}\{u/k\}u^{-2} du$ and $\langle \chi, \varsigma_k \rangle = \int_1^\infty \{u/k\}u^{-2} du$. On an interval (a, b) on which $\{u/j\} = u/j - \alpha$ and $\{u/k\} = u/k - \beta'$ with integers α, β' ,*

$$\int_a^b \frac{\{u/j\}\{u/k\}}{u^2} du = \frac{b-a}{jk} - \left(\frac{\alpha}{k} + \frac{\beta'}{j} \right) \log \frac{b}{a} + \alpha\beta' \left(\frac{1}{a} - \frac{1}{b} \right).$$

Over a period of $\{u/j\}\{u/k\}$ the mean is $\frac{1}{4} + \gcd(j, k)^2/(12jk)$.

Proof. The first two formulas are the substitution $u = 1/t$; the third is the integral of $1/(jk) - (\alpha/k + \beta'/j)u^{-1} + \alpha\beta'u^{-2}$. For the mean, write $\{y\} = \frac{1}{2} + B(y)$ with $B(y) = -\sum_{n \geq 1} \sin(2\pi ny)/(\pi n)$, so that the mean of $\{u/j\}\{u/k\}$ is $\frac{1}{4}$ plus the mean of $B(u/j)B(u/k)$. With $g = \gcd(j, k)$, $j = gj'$, $k = gk'$, the products of sines have nonzero mean only for $n/j = m/k$, that is $n = j't$, $m = k't$, and the mean is $\sum_{t \geq 1} 1/(2\pi^2 j'k't^2) = 1/(12j'k') = g^2/(12jk)$. $\square$

N	d_N^2	$d_N^2 \log N$	N	d_N^2	$d_N^2 \log N$
1	0.858212	–	12	0.020443	0.0508
2	0.289965	0.2010	15	0.018335	0.0497
3	0.093837	0.1031	20	0.016504	0.0494
4	0.065129	0.0903	25	0.015520	0.0500
5	0.036102	0.0581	30	0.014530	0.0494
6	0.035986	0.0645	40	0.012715	0.0469
8	0.024161	0.0502	50	0.011869	0.0464
10	0.023772	0.0547	60	0.011471	0.0470

Table 3: The sawtooth distance from exact integrals of fractional parts.

Computation. For $j, k \leq 60$ we integrate exactly, by Theorem 6.1, on $(0, U]$ with U the least multiple of $\text{lcm}(j, k)$ above $4 \cdot 10^6$, split at the multiples of j and of k , and add the tail $(\frac{1}{4} + \gcd(j, k)^2/(12jk))/U$; the error of the tail is below $\text{lcm}(j, k)/U^2 < 3 \times 10^{-10}$. On intervals (a, b) with $a \geq 2 \cdot 10^4$ the closed form of Theorem 6.1 cancels to many digits, and we use instead its expansion in powers of $1/a$, $jk \int_a^b \{u/j\}\{u/k\}u^{-2} du = \int_0^{b-a} (r + \alpha_0)(r + \beta_0)(a + r)^{-2} dr = \sum_{n \geq 0} (-1)^n (n+1) a^{-n-2} \int_0^{b-a} (r + \alpha_0)(r + \beta_0) r^n dr$ with $\alpha_0 = j\{a/j\}$, $\beta_0 = k\{a/k\}$, truncated

after seven terms ($(b-a)/a \leq 3 \times 10^{-3}$). The inner products $\langle \chi, \varsigma_k \rangle$ are computed in the same way with the tail $\frac{1}{2}/U$, and agree with the closed form $(\log k + 1 - \gamma)/k$ to 4×10^{-13} . The checks $\langle \varsigma_1, \varsigma_1 \rangle = \log 2\pi - \gamma = 1.2606614015078$ and $\langle \chi, \varsigma_1 \rangle = 1 - \gamma = 0.4227843350985$ hold to all 13 printed digits, and $\langle \varsigma_2, \varsigma_2 \rangle = \frac{1}{2} \langle \varsigma_1, \varsigma_1 \rangle$ as the scaling $\varsigma_k(t) = \varsigma_1(kt)$ requires. Then $d_N^2 = 1 - b^\top G^{-1} b$ with G the Gram matrix and b the vector of $\langle \chi, \varsigma_k \rangle$; the condition number of G is 1.3×10^4 at $N = 60$. Table 3 and Fig. 2b give the values. For $40 \leq N \leq 60$, $d_N^2 \log N$ lies between 0.0454 ($N = 42$) and 0.0470 ($N = 58$ and 60), within 1.8% of $2 + \gamma - \log 4\pi = 0.0461914$, and crosses it between $N = 48$ and $N = 49$: the distance by which N sawtooths of the exact count fall short of the constant, per unit of $1/\log N$, agrees in this range with the energy per unit of $\log x$ of the prime count's error in Section 5, and under RH and the hypothesis of [3] the two are equal in the limit.

7 The rebuild law beyond 10^{10}

With a table of μ to $L = 10^9$, built by the pairing (each prime flips the sign of its multiples and each prime square annihilates them), Theorem 2.1 evaluates $M(x)$ from the law and the table. It returns $M(10^9) = -222$, $M(10^{10}) = -33,722$, $M(10^{11}) = -87,856$, $M(10^{12}) = 62,366$, $M(10^{13}) = 599,582$, $M(10^{14}) = -875,575$ and $M(10^{15}) = -3,216,373$, the published values [13]; $M(10^{10})$ also agrees with our sieve. Table 4 gives the doublings, which agree with [14].

x	$M(x)$	$M/\sqrt{x}$	x	$M(x)$	$M/\sqrt{x}$
2^{34}	-3,421	-0.026	2^{42}	-169,338	-0.081
2^{35}	8,435	0.046	2^{43}	259,886	0.088
2^{36}	38,176	0.146	2^{44}	-474,483	-0.113
2^{37}	-28,118	-0.076	2^{45}	1,726,370	0.291
2^{38}	38,729	0.074	2^{46}	-3,554,573	-0.424
2^{39}	-135,944	-0.183	2^{47}	-135,443	-0.011
2^{40}	101,597	0.097	2^{48}	3,282,200	0.196
2^{41}	15,295	0.010	2^{49}	1,958,235	0.083

Table 4: $M(x)$ from the rebuild law at the doublings 2^{34} to 2^{49} .

At $10^{11}, \dots, 10^{15}$ the ratio $M/\sqrt{x}$ is $-0.278, 0.062, 0.190, -0.088$ and -0.102 . Over all the points evaluated the largest value of $|M|/\sqrt{x}$ is 0.424, at 2^{46} .

8 The small primes and the newest primes

The integers free of the primes up to 37. Let $P = \prod_{p \leq 37} p$ and $\Phi_{37}(x) = \#\{n \leq x : (n, P) = 1\} = \sum_{d|P} \mu(d) \lfloor x/d \rfloor$. Then $\Phi_{37}(x) - x \prod_{p \leq 37} (1 - 1/p) = -\sum_{d|P} \mu(d) \{x/d\}$ is bounded by 2^{11} and periodic with period P . Evaluated exactly on the grid of Section 4, its absolute value is at most 38.9 for $2^{12} \leq x \leq 10^{10}$, and at $x = 10^{10}$ it is $18.4 = 0.00018\sqrt{x}$; its spectrum in doublings has amplitude at most 0.0018 at every pitch in $[0.3, 6]$, and at most 0.0017 at the pitches of Table 2, against 0.14–0.18 for the first wave of the prime counts. Of the 1,487,210,140 integers up to 10^{10} that it counts, 455,052,499 are prime. Since the primes up to 37 contribute $O(\log x)$ to $\psi(x)$ and the sieve by them leaves an error bounded by 2^{11} , the waves of Section 4 lie in the placement of the primes above 37 among the integers free of the primes up to 37.

The newest primes. Let $M(x, y) = \sum_{n \leq x, P^+(n) \leq y} \mu(n)$, $P^+(n)$ the largest prime factor. An integer $n \leq x$ with $P^+(n) = p > x/2$ is $n = p$, so

$$M(x) = M(x, x/2) - (\pi(x) - \pi(x/2)).$$

At $x = 10^{10}$, $\pi(x) = 455,052,511$, $\pi(x/2) = 234,954,223$ and $M(x) = -33,722$, so $M(x, x/2) = 220,064,566 = 2200.6\sqrt{x}$: the primes in $(x/2, x]$ bring the balance from $2200.6\sqrt{x}$ to $-0.337\sqrt{x}$. The sieve also gives $M(x, \sqrt{x}) = -219.4\sqrt{x}$ at $x = 10^{10}$. At $x = 10^{10}$ the sum reaches the scale $\sqrt{x}$ only when the primes in $(x/2, x]$ are included.

A sign that holds for 9×10^8 steps. The sum $L(x)$ of the Liouville function satisfies $L(x) \leq 0$ for $2 \leq x < 906,150,257$ and $L(906,150,257) = 1$ [17], which our sieve confirms.

Remark 8.1 (the count must be exact). In the system of Theorem 1.4 and in the systems of [5] the pairing of Theorem 1.1(i) holds, because factorisation into generalized primes is unique, and so does the rebuild law over the generalized integers. In the system of Theorem 1.4, and in the systems of [5] with $\alpha > \frac{2}{3}$ below, the energy $E_B(X) = \int_1^X ((\psi_B(x) - x)/x)^2 dx$ exceeds a fixed positive power of X for arbitrarily large X . Indeed ψ_B is nondecreasing, so if $0 < h \leq x_0$ and $\psi_B(x_0) - x_0 \geq h$, then $\psi_B(x) - x \geq h/2$ on $[x_0, x_0 + h/2]$, and if $\psi_B(x_0) - x_0 \leq -h$, then $\psi_B(x) - x \leq -h/2$ on $[x_0 - h/2, x_0]$; either interval contributes at least $h^3/(18x_0^2)$ to $E_B(2x_0)$. With $h = x_0^{\alpha'}$ this is $x_0^{3\alpha'-2}/18$. In the system of Theorem 1.4, let $\frac{1}{2} < \alpha' < 1$. If $\psi_B(x) - x = O(x^{\alpha'})$, then, since the prime powers $p^k \leq x$ with $k \geq 2$ contribute $O(\sqrt{x} \log x)$ to $\psi_B(x)$, the sum $\theta_B(x)$ of $\log p$ over the generalized primes $p \leq x$ satisfies $\theta_B(x) - x = O(x^{\alpha'})$, and partial summation gives $\pi_B(x) - \text{li}(x) = O(x^{\alpha'})$, contrary to $\pi_B(x) - \text{li}(x) = \Omega(x \exp(-c\sqrt{\log x}))$; so $|\psi_B(x) - x| \geq x^{\alpha'}$ for arbitrarily large x . Broucke, Debruyne and Révész [5] construct, for each $\alpha \in [0, 1)$ and $\beta \in [\frac{1}{2}, 1)$, systems with $\psi_B(x) = x + O(x^{\alpha+\varepsilon})$ and $N_B(x) = \kappa x + O(x^{\beta+\varepsilon})$ for every $\varepsilon > 0$, neither exponent being improvable; with $\beta = \frac{1}{2}$ and $\alpha > \frac{2}{3}$, $|\psi_B(x) - x| \geq x^{\alpha'}$ for arbitrarily large x whenever $\alpha' < \alpha$, and taking $\frac{2}{3} < \alpha' < \alpha$ the energy exceeds a power of X for arbitrarily large X . An argument that derived Conjecture 1 from the pairing, the rebuild law and a bound $O(x^{1/2+\varepsilon})$ for the error of the integer count would apply to these systems as well, where its conclusion is false. For the ordinary integers the error of the count is $-\{x\}$, bounded by 1, and this is what d_N measures: the sawtooths ς_k are the errors of the counts $\lfloor 1/(kt) \rfloor$.

9 Methods of computation

Counts to 10^{10} . An exact segmented sieve to 10^{10} gives $\mu(n)$, $\lambda(n)$ and $\Lambda(n)$ and records M , L , π and $\psi - x$ in extended precision at 675,247 checkpoints; at the top of the range consecutive checkpoints differ by 3.3×10^{-4} in $\log_2 x$. It returns $\pi(10^{10}) = 455,052,511$, $M(10^{10}) = -33,722$ and $L(10^{10}) = -116,026$.

The rebuild law. Theorem 2.1 in 64-bit integer arithmetic with $L = 10^9$; $M(10^{15})$ takes about 10^{11} integer divisions.

Spectra and energy. Linear interpolation of the checkpoint values on the grid $\Delta u = 10^{-3}$, Hann window, direct evaluation of the Fourier sum at each pitch.

Zeros. The 2469 zeros of ζ with $0 < \gamma < 3000$ are the sign changes of Hardy's function $Z(t)$, evaluated by Euler–Maclaurin summation on a grid of one twelfth of the mean spacing and refined by Brent's method to 10^{-13} [16]; an independent evaluation of Z by the Riemann–Siegel formula gives the same 2469 sign changes and the same $\sum 2/(\frac{1}{4} + \gamma^2)$ to six digits. All lie on the critical line [15].

Sawtooth integrals. Theorem 6.1 in double precision; the linear systems by Gaussian elimination.

The count of Section 8. Inclusion–exclusion over the 4096 squarefree divisors of $\prod_{p \leq 37} p$ in exact integer arithmetic.

10 Concluding remarks

What the paper fixes about the remaining step. The pairing makes the rebuild law hold, and the law fixes the balance at every scale from the smaller ones (Theorem 1.1); the pair-weighted sawtooths rebuild the constant exactly, and every finite rebuild falls short by the value imbalance and by the unfinished pairs (Theorem 1.2); RH is the subpolynomial growth of the energy of the count’s error (Theorem 1.3), and the energy measured in the counts, the energy of the waves and the sawtooth distance per unit of $1/\log N$ agree with the single constant $2 + \gamma - \log 4\pi$ to within 2%. The pairing and the law hold in Beurling systems whose energy exceeds a power of X for arbitrarily large X (Theorem 8.1), so the step must use more about the count than these; the ordinary count is exact.

Outlook. The next computations are the energy of M and of ψ in doublings beyond 10^{10} , from the rebuild law and its analogue for ψ , and d_N for N in the thousands by the same exact integrals. The next analytic question is how the exactness of the count, the bound $|\{x\}| < 1$ that no Beurling system with a power-size error shares, enters a bound for $E(X)$.

11 Data and sources

External inputs. All inputs are mathematical; no measured data are used.

- (i) All nontrivial zeros of ζ with $0 < \gamma \leq 3 \cdot 10^{12}$ lie on the critical line [15].
- (ii) Published values used for comparison: $M(10^n)$, $n \leq 15$ [13]; $M(2^n)$, $34 \leq n \leq 49$ [14]; $\pi(10^{10}) = 455,052,511$ [12]; the least $x > 1$ with $L(x) > 0$, 906,150,257 [17].
- (iii) Classical and published theorems: Littlewood’s criterion [18, §14.25]; von Koch’s bound [20]; Cramér’s mean value theorem [7]; the explicit formula and $\sum_{\rho} \operatorname{Re}(1/\rho)$ [9]; $\sum \mu(k)/k = 0$ [1]; the criterion of Nyman, Beurling and Báez-Duarte [4, 2]; Burnol’s bound [6]; the asymptotic of Bettin, Conrey and Farmer [3]; the Beurling systems of [10, 5]; the closed forms of the Gram integrals [19] and the earlier computations of d_N [8]; $\limsup |M(x)|/\sqrt{x} > 1$ [11].

Computed quantities. Every other number in the paper is computed from formulas stated in the text by the methods of Section 9: the counts to 10^{10} , the spectra, growth exponents and energies, the sums over zeros, the Gram matrices and d_N , the values of M from the rebuild law to 10^{15} , $m(N)$ to 10^8 , and the count of integers free of the primes up to 37.

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